Water Distribution
Anomaly Detection & Quality of Service Report

Benchmarking, Pattern Analysis, and Daily Anomaly Scoring

Data Period: 05 Aug 2026 | Benchmark: Hardcoded fallback (DB building up)

Data Source: VMC.DLP3.JAM.AIB_FT015 (Flow Rate) | Meter: AIB_FT015

Generated: 05 Aug 2026, 23:06 IST

QoS Score: 44.0% | Status: POOR - ACTION REQUIRED

FLOW STATUS	ANOMALY RATE	SUPPLY HOURS
471 m³/hr Peak: 960 m³/hr	**23.6%** 56 flags	**23.8 hrs** 1 window(s)

1. Executive Summary

This report covers water distribution data for **05 Aug 2026**. A total of **237** readings were analysed from meter **AIB_FT015**. The overall Quality of Service score is **44.0%** with **56** anomalies detected (23.6% of readings). The 4-model ensemble (Z-score: 0, IQR: 0, Isolation Forest: 12, PCA: 2) produced 9 final anomaly flags.

Metric	Value	Metric	Value
Report Date	05 Aug 2026	**QoS Score**	44.0%
Total Readings	237	**Status**	POOR - ACTION REQUIRED
Average Flow	470.5 m³/hr	**Total Anomalies**	56
Peak Flow	960.0 m³/hr at 04:37	**Spike Anomalies**	0
Supply Windows	1	**Night Anomalies**	56
Benchmark Samples	1	**Z-score Anomalies**	0

2. Benchmark Profile vs Today

The benchmark was constructed from Jan–Feb supply window data. Today's detected sessions are compared below.

Session	Start Time	End Time	Duration (min)	Peak Flow (m³/hr)	Avg Flow (m³/hr)	Source
Benchmark	00:00	23:54	1380	5500.0	1000.0	Jan–Feb (1 days)
Today #1	23:11	23:02	1431	960.0	470.5	Today

3. Benchmark Parameters

The benchmark is derived from **1** historical supply windows. The primary supply session begins around **00:00** and ends around **23:54**, lasting approximately **1380 minutes** with a peak flow of **5500.0 m³/hr**.

Parameter	Benchmark Value	Tolerance
Start Time	00:00	± 30 min
End Time	23:54	± 30 min
Duration	1380 min	± 30 min
Peak Flow	5500.0 m³/hr	± 20%
Average Flow	1000.0 m³/hr	± 20%
Sample Size	1	—

3. Today's Supply Windows

#	Start	End	Duration	Peak Flow	Avg Flow	Status
1	23:11	23:02	1431 min	960.0 m³/hr	470.5 m³/hr	Deviated

4. Anomaly Details

#	Description	Severity
1	Start time off by 1391 min (benchmark: 00:00)	High
2	Peak flow deviated by 83% (benchmark: 5500.0 m³/hr)	High
3	Avg flow deviated by 53% (benchmark: 1000.0 m³/hr)	High

5. Multi-Model Anomaly Analysis

The following table shows anomaly counts per detection method from the 4-model ensemble applied to today's batch data.

Method	Anomalies Flagged	Description
Z-score	0	Supply-hour readings > 3σ from mean
IQR	0	Beyond Q1 – 2.5×IQR or Q3 + 2.5×IQR fence
Isolation Forest	12	Tree-based outlier score (contamination=5%)
PCA Autoencoder	2	Reconstruction error > mean + 3σ
Final (3+ / rule)	9	≥3 models agree, or rule-based (spike / night / drop)

6. Top Anomalous Readings

#	Timestamp	Flow Rate (m³/hr)	Type
1	04:37:16	960.0	Night
2	04:36:59	930.0	Night
3	04:37:29	930.0	Night
4	03:00:51	912.0	Night
5	03:44:29	900.0	Night
6	03:27:17	900.0	Night
7	03:08:11	900.0	Night
8	02:55:58	900.0	Night
9	04:02:11	900.0	Night
10	04:41:33	900.0	Night

7. Daily Anomaly Detail

The following 3 anomaly/anomalies were detected today against the Jan–Feb benchmark.

#	Anomaly Description	Category
1	Start time off by 1391 min (benchmark: 00:00)	Timing
2	Peak flow deviated by 83% (benchmark: 5500.0 m³/hr)	Flow Rate
3	Avg flow deviated by 53% (benchmark: 1000.0 m³/hr)	Flow Rate

8. Flow Forecast

Exponential smoothing (α=0.3) forecast for the next **30 readings**. Predicted range: **2.8 – 1058.4 m³/hr** (95% CI). Trend direction: Decreasing.

Step	Forecast (m³/hr)	Lower 95%	Upper 95%
1	530.6	434.2	626.9
2	530.6	394.3	666.8
3	530.6	363.7	697.5
4	530.6	337.9	723.3
5	530.6	315.1	746.0
6	530.6	294.5	766.6
7	530.6	275.6	785.5
8	530.6	258.0	803.1
9	530.6	241.5	819.7
10	530.6	225.9	835.3

10. Pattern Analysis (Jan–Feb 2026)

K-Means clustering (k=6) was applied to Jan–Feb 2026 daily flow curves. The modal cluster centroid was set as the benchmark pattern. Days within 75% similarity are marked as matching.

Metric	Value
Total days analysed	59
January days	31
February days	28
Matching days (≥threshold)	0
Deviant days (<threshold)	59
Average similarity	0.0%
Best matching day	01-01
Worst matching day	01-01
Modal cluster index	1
Similarity threshold	75%

Figure 1. Multi-Day Overlay — Jan–Feb 2026 (each line = 1 day, red = median)

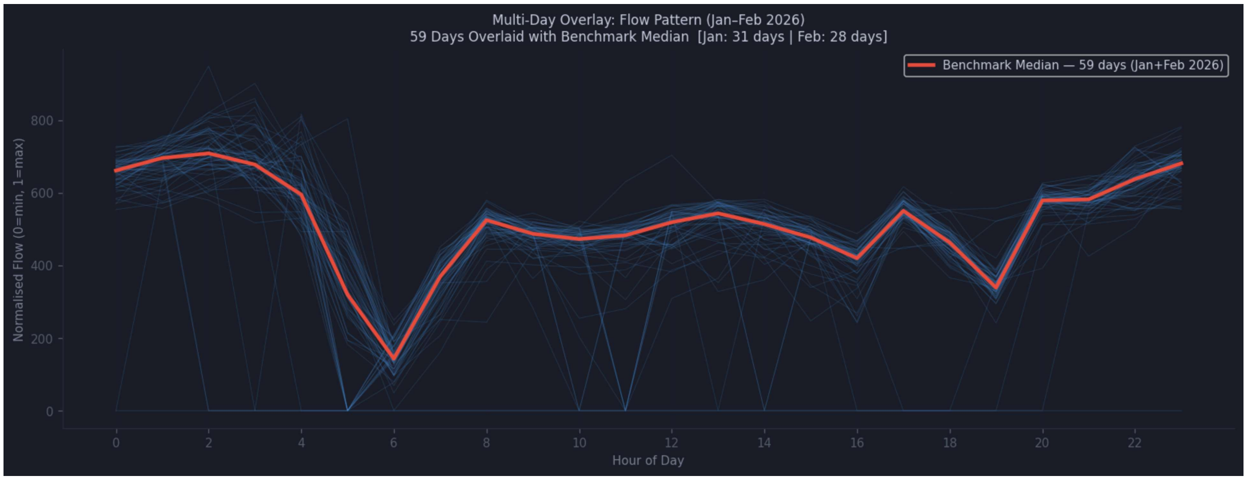


Figure 2. K-Means Cluster Centroids (k=6) — orange = benchmark cluster

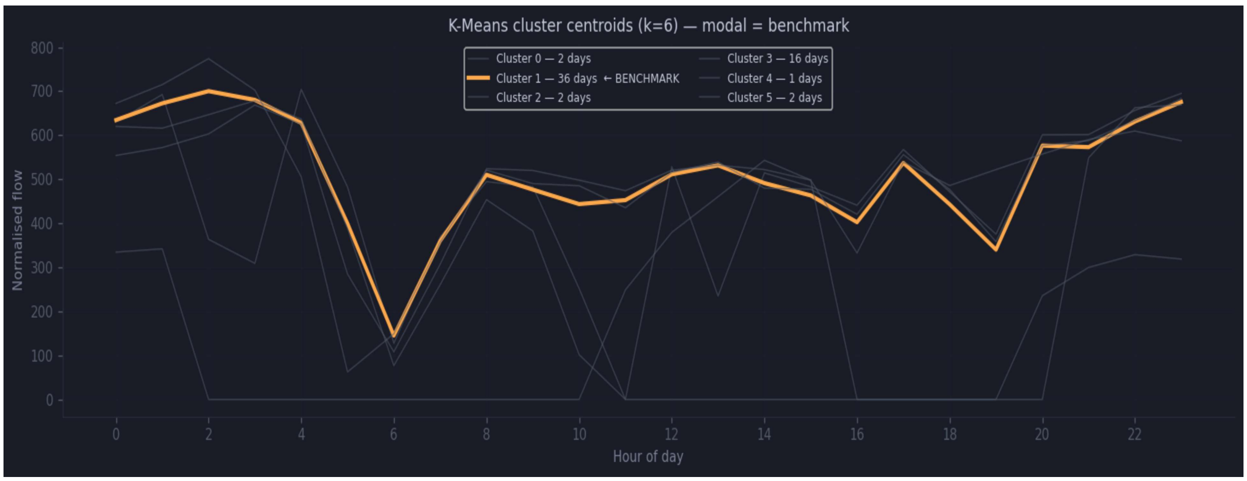


Figure 3. Daily Pattern Similarity — green ≥ 75%, red = deviant

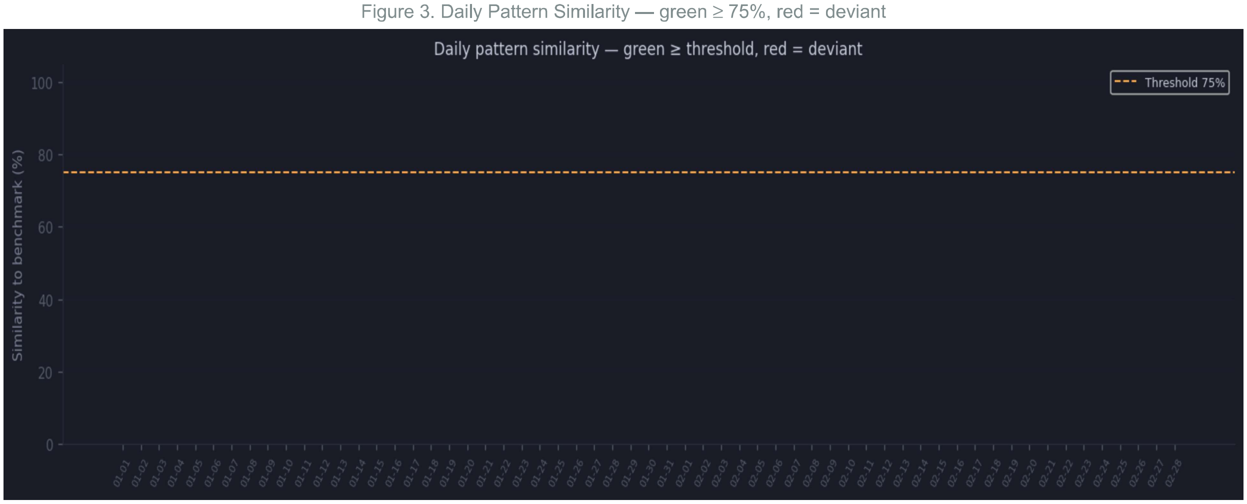
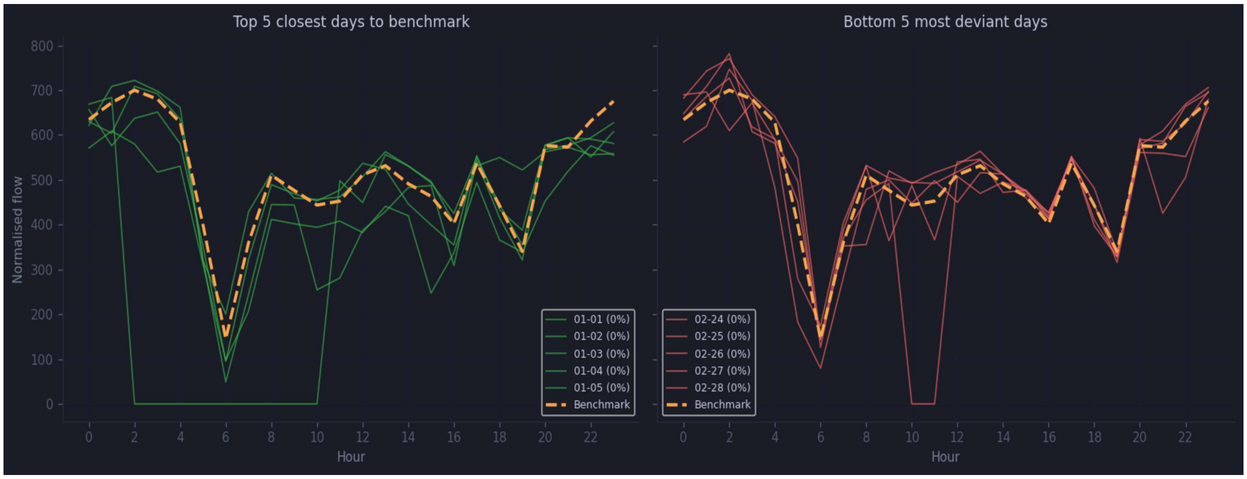
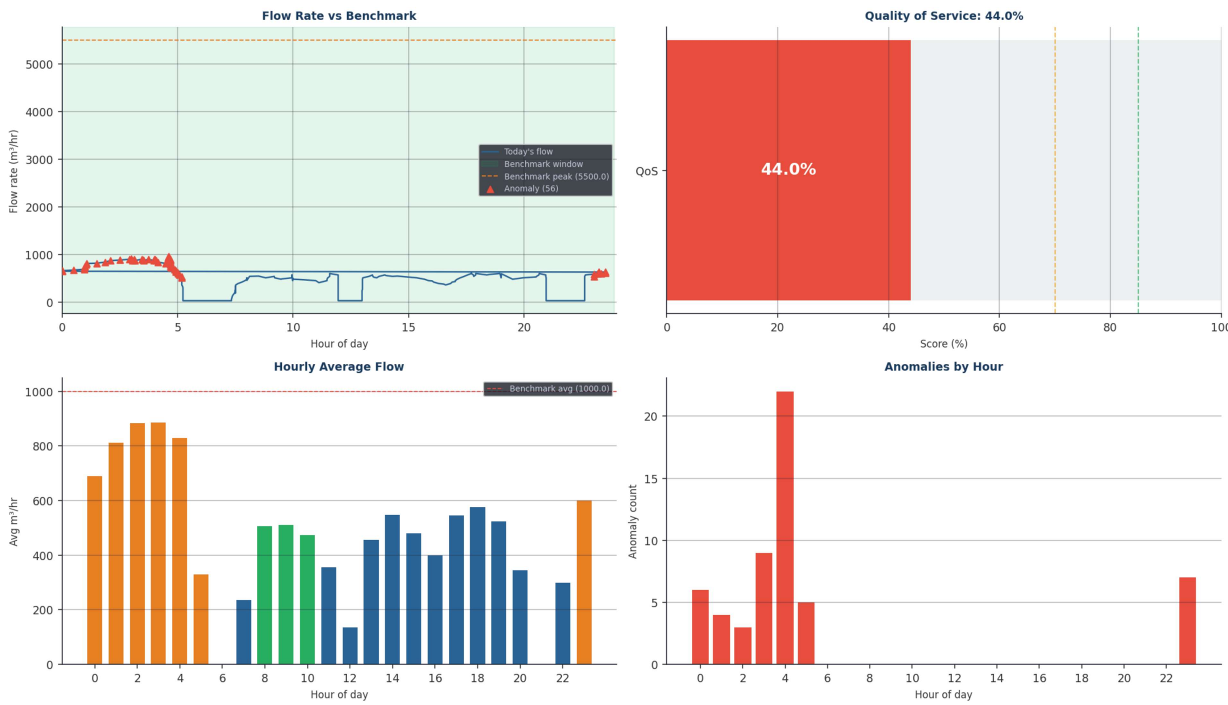


Figure 4. Best-match vs Worst-match Days vs Benchmark



11. Flow Analysis Charts



12. Recommendations

Category	Recommendation
Scheduling	Supply start time is deviating from schedule. Check pump station readiness and valve operations.
Flow Rate	Flow rate deviation detected. Inspect pump performance and pipe conditions.
Night Flow	56 readings above threshold during non-supply hours. Check for unauthorized connections or leakage.
Pattern Drift	59 deviant days found in Jan–Feb 2026. Benchmark may need updating to reflect seasonal or infrastructure changes.
Flow Rate Monitoring	Average flow (470.5 m³/hr) is >20% below benchmark (1000.0 m³/hr). Inspect pump performance and check for pipe blockag
Reverse Flow	Monitor non-supply-hour flow. Any sustained readings >200 units during off-hours should trigger investigation for unauthorize
Benchmark	Update benchmark monthly using a rolling 60-day window to capture seasonal demand variations.

13. 7-Day QoS Trend

Date	QoS%	Avg Flow	Peak Flow	Anomalies	Spikes	Status
2026-08-05	44.0%	471	960	55	0	POOR
2026-07-29	43.0%	441	846	53	0	POOR
2026-07-28	41.1%	461	846	41	0	POOR

14. Today's Service Classification

Classification	Value	Description
Day Category	Critical day (3 anomalies)	Perfect=0 anomalies, Minor=1-2, Critical=3+
QoS Band	Poor (<70%)	Excellent≥85%, Good 70-85%, Poor<70%
Anomaly Count	3	Total supply-window deviations from benchmark
Supply Windows	1	Active supply sessions detected today
Benchmark Days	1	Jan–Feb days used to build benchmark